Similarity Self-Check Report

Cortex: An Agentic Retrieval-Augmented Generation System over Live Work Data with Explainable Memory and Consent-Gated Actions

Self-check, not an official certificate. The project team produced this report with open tools and published sources. It is not a Turnitin, DrillBit/ShodhShuddhi, iThenticate or Urkund report, and it did not compare the text against any repository of student papers. Use it to find passages to revise before the official institutional check.

Similarity index (external sources)

0.1%

18 of 16571 checked words lie inside 8-word sequences found in a cited source or on the web

Excluding titles of cited works and venue names: **0.0%**

Document	docs/report/report.pdf (BE project report, 128 pages)
Authors	Ajit Paraskar, Prathamesh Kumbhar, Harshvardhan Patil, Bikram Ghosh; guide Prof. Swapnali Gaikwad (Dr. D. Y. Patil Institute of Technology, Pimpri)
Checked on	31 October 2026
Words checked	16571 (after exclusions below)
Match unit	identical sequence of 8 or more words (case and punctuation ignored)
Self-overlap	0.2% of words also appear in the team's own paper (same authors and project). Reported separately and **not** counted in the index.

48
unique cited references: 39 compared as arXiv full text, 0 by abstract, 9 by title only

6
sources with at least one 8-word match

43
exact-phrase web searches run for this document

0
web searches that found the phrase elsewhere

Top sources

#	Source	Text compared	Words	Share
1	Search-o1: Agentic Search-Enhanced Large Reasoning Models https://arxiv.org/abs/2501.05366	arXiv full text	10	0.06%
2	Dense X Retrieval: What Retrieval Granularity Should We Use? https://arxiv.org/abs/2312.06648	arXiv full text	10	0.06%
3	RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval https://arxiv.org/abs/2401.18059	arXiv full text	10	0.06%
4	Dense Passage Retrieval for Open-Domain Question Answering https://arxiv.org/abs/2004.04906	arXiv full text	10	0.06%
5	FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation https://arxiv.org/abs/2305.14251	arXiv full text	10	0.06%
6	GRASP: GRanularity-Aware Search Policy for Agentic RAG https://arxiv.org/abs/2607.10463	arXiv full text	8	0.05%

Share = document words inside matched sequences from that source ÷ words checked. A word matched by two sources counts once in the index.

What was checked and what was left out

- Excluded: Front matter pp.1-4 and 6-12: cover, certificate, declaration, acknowledgement, table of contents, lists of figures and tables
- Excluded: Chapter 13 References and Annexure A References (pp.99-106)

- Excluded: Annexure F Plagiarism Report placeholder (pp.119-120)
- Excluded: Annexure H group-member information and back matter (pp.123-128)
- Excluded: Algorithm pseudo-code lines (lines with assignment/set symbols or control keywords) and monospace code spans
- Excluded: Running page headers/footers and page numbers
- **Source check (whole text):** every cited reference in refs.bib / biblo.bib was looked up in the Semantic Scholar Graph API; where an arXiv id existed, the full paper text was downloaded from arXiv. Every 8-word sequence of the document was compared with every source.
- **Web check (sample):** a stratified random sample of 253 sentences of 10 or more words, drawn per chapter with seed 2026; 50 of them (random, seed 2026) were also searched on the web. For each, a distinctive 10-word span was searched as an exact phrase with Claude's WebSearch tool.
- **Cross-document check:** 8-word overlap with the team's own paper.

Matched passages (external sources)

#	Page	Section	Words	Matched text	Assessment
1 +2,3,4,5	18	Synopsis	10	*"acm sigir conference on research and development in information retrieval"*	venue name (not plagiarism)
6	5	Abstract	8	*"grasp granularity aware search policy for agentic rag"*	title of the cited work (not plagiarism)

Paragraphs that contain an external match

Coloured text with a number = matched an external source (number = row in Top sources).

PAGE 5 · ABSTRACT

The project's research framing follows recent work on agentic RAG, in particular GRASP (GRanularity-Aware Search Policy for Agentic RAG[6]), which studies when and how finely an agent should retrieve. In a preliminary 12question evaluation over a fictional sample workspace, using free-tier models, Cortex reached a mean groundedness of 0.88 and a citation recall of 0.92, answered the unanswerable question honestly, executed no action without consent, and had a median latency of 3.7 s (p90 7.1 s). Conflict detection and action proposals each succeeded in one of two cases, which identifies the main directions for further work.

PAGE 18 · SYNOPSIS

• ACM SIGIR Conference on Research and Development in Information[1] Retrieval[1] (ACM).

Self-overlap with the team's own paper

33 words (0.2%) of this document also occur in 8-word sequences of the paper. Both documents describe the same project and were written by the same team, so this is expected. If both are submitted to the same checker, the second one checked may show this overlap; ask the coordinator to exclude the team's own submission or declare it. Longest shared passages:

Words	Page	Passage
9	51	*"gmail google calendar google drive notion slack and github"*
8	33	*"run in parallel with a 12 s timeout"*
8	5	*"project this week and what is at risk"*
8	17	*"at most 5 sub questions 18 tool calls"*

Web exact-phrase searches

A "none" result means no page in the results contained the phrase. Topical results (pages about the same subject, e.g. the cited paper itself) are noted but are not matches. Calibration searches below show how reliable this search is.

Sent.	Page	Phrase searched	Result
2	5	*"risk?" means opening each tool and reconciling what they say"*	none
9	5	*"free tiers through a failover router across several language model"*	none
110	17	*"Functional relations: F = {fplan, fsearch, frank, fsynth, fverify, fpropose"*	skipped:non-prose
137	19	*"starting from short quotes and capping how many retrieval turns"*	none
141	19	*"Cortex keeps the core idea of answering from retrieved text"*	none
148	19	*"The authors fire a new retrieval after every reasoning sentence"*	none (topical: IRCoT 2212.10509)
164	19	*"A classifier guesses how hard a question is and picks"*	none
296	27	*"learned behaviour, the authors see the agent search by meaning"*	none
314	28	*"RARR takes finished model output, searches for support, and edits"*	none (topical: RARR 2210.08726, cited)
333	29	*"map the MCP ecosystem and list the attacks possible at"*	none (topical: MCP 2503.23278, cited)
369	31	*"question q from this user, Cortex has a fixed allowance"*	not searched
421	33	*"same search repeated within s comes from a separate search"*	not searched
453	35	*"□ A developer can check which pull requests wait on"*	none
455	35	*"□ Anyone who starts the day with e-mail can read"*	none
488	39	*"The estimates below use the basic COCOMO model together with"*	none
506	39	*"□ Hardware: the team's laptops (Table no dedicated server is"*	not searched

540	41	*"Are end-users enthusiastically committed to the project and the system/product"*	none (note: sentence is a question from Pressman & Maxim's risk-item checklist; cited in report as [1])
554	41	*"The functional requirements are stable; external APIs and free-tier limits"*	none
654	47	*"□ Task Feasibility spikes: free-tier model failover, Composio OAuth"*	none
698	49	*"Composio, MCP, Firecrawl and sample connectors; freetier router, cooldowns and"*	skipped:list-of-names
705	49	*"External guide Industry guidance on behalf of the sponsor, Techtrix"*	none
808	55	*"source, title, quote (a compact excerpt"*	skipped:field-list
820	55	*"connection expired; reconnect it") and the impact on the answer"*	none
868	56	*"response cache and the search cache are omitted for clarity"*	none
976	60	*"means writing one class that implements the connector interface, or"*	none
983	60	*"only once, and only for the user it was proposed"*	none
1019	61	*"time; a restart discards them, which is safe (nothing executes"*	none
1048	62	*"LangSmith API Optional traces of runs, model calls and searches"*	none
1061	64	*"browser client (React) renders the Ask, Brief, Memory and Connections"*	none
1070	64	*"cache. stands on its own and is reached over HTTP"*	none
1080	65	*"Ask + live trace Brief Memory inspector Connections useAgent (SSE"*	skipped:non-prose
1153	67	*"□ Prompts: the planning prompt (returns a JSON plan), the"*	none
1252	73	*"algorithm was written from the actual code in the repository"*	none
1279	75	*"emit a failure event (source, reason, impact) for each unavailable"*	none
1354	82	*"The evaluation used Deep mode over the fictional sample workspace"*	none
1367	83	*"JSON inside a code fence, or embedded in prose"*	none
1397	84	*"Remember that Orbit is my top priority"*	skipped:example-user-utterance
1438	87	*"interface review with the development mock model, which quotes evidence"*	none
1439	87	*"Figure Ask screen: a cited answer with a struck-through unsupported"*	none
1447	87	*"The lowest values and came from questions about deadlines ("oa-"*	skipped:numeric-fragment
1451	90	*"Figure Memory inspector (empty state) with the control to teach"*	none
1543	94	*"Un-installation: stop both processes; delete the repository folder (this removes"*	none
1556	95	*"Select Trace to see the plan, the searches and their"*	none

1581	97	*"Over twelve questions the preliminary results back the main design"*	none
1595	97	*"□ Implement the remaining vision items: time-travel queries over source"*	none
1627	109	*"Increase Trust in answers: every factual sentence is cited and"*	none
1642	110	*"Knowledge sources Gmail, Calendar, Drive, Notion, Slack, GitHub, MCP servers"*	skipped:list-of-names
1708	113	*"The references are kept for completeness; the McKinsey reports are"*	none
1729	116	*"The project schedule is planned with a Gantt chart (Figure"*	none (generic wording; no exact match)
1773	122	*"competitions/conferences participated like IIT, Central Universities, University Conferences or equivalent"*	covered-by-probe

Calibration and targeted probes

Probe	Result
control1	found:https://arxiv.org/pdf/2005.11401:we introduce RAG models where the parametric memory is a pre-trained seq2seq model
control2	missed:Lewis et al. 2020 abstract opening 'pre-trained language models have been shown to store factual knowledge in their parameters' (true source arXiv 2005.11401 not returned)
control3	missed:Pressman risk questionnaire 'Have top software and customer managers formally committed to support the project' (textbook text; not returned)
control4	missed:SPPU template 'To develop the problem under consideration and justify feasibilty using concepts of knowledge canvas'
probe1607	none (SPPU syllabus heading 'justify feasibilty using concepts of knowledge canvas and IDEA Matrix')
probe1617	none (SPPU syllabus heading 'satisfy ability issues using modern algebra')
probe1782	none (SPPU syllabus heading 'generate various testing result charts')
probe1758	none (SPPU template 'conferences/workshops in IITs, Central Universities or UoP')

Appendix: sentence check set

A stratified random sample of 253 sentences of 10 or more words, drawn per chapter with seed 2026; 50 of them (random, seed 2026) were also searched on the web. Sentence numbers refer to the order of sentences in the extracted text (raw list in report_sentences.json). Every sentence in the document, not only these, went through the source check.

Sent.	Page	Section	Sentence	Checks
2	5	Abstract	*Answering "what changed in our project this week, and what is at risk?" means opening each tool and reconciling what they say.*	web + source
6	5	Abstract	*Each claim is then checked against the evidence it cites and a groundedness score is computed; unsupported claims are struck through, and contradictions between sources a…*	source
9	5	Abstract	*The system runs on free tiers through a failover router across several language model providers.*	web + source
10	5	Abstract	*The project's research framing follows recent work on agentic RAG, in particular GRASP (GRanularity-Aware Search Policy for Agentic RAG), which studies when and how finel…*	source
12	5	Abstract	*Conflict detection and action proposals each succeeded in one of two cases, which identifies the main directions for further work.*	source
15	14	Synopsis	*Cortex: An Agentic Retrieval-Augmented Generation System over Live Work Data with Explainable Memory and Consent-Gated Actions*	source
26	14	Synopsis	*The keywords below follow the 1998 ACM Computing Classification System ().*	source
61	15	Synopsis	*Every factual claim in its answer is cited and then checked against the evidence it cites.*	source
68	15	Synopsis	*A typical RAG chatbot runs one similarity search over a fixed corpus and treats every retrieved sentence as equally reliable.*	source
71	15	Synopsis	*For each question it plans two to five sub-questions, searches the connected sources in parallel through a registry of interchangeable connectors, recalls the priorities …*	source
73	16	Synopsis	*exist, the content words must overlap and the numbers must match.*	source
75	16	Synopsis	*The user sees a groundedness score and a confidence state for each claim.*	source
82	16	Synopsis	□ *Answer multi-part questions over several live sources with a bounded*	source
86	16	Synopsis	□ *Surface cross-source contradictions as risks with both citations, and*	source
88	16	Synopsis	□ *Never execute an action in a user's tool without an explicit, single-use*	source
92	16	Synopsis	□ *Degrade gracefully: name any source that fails, explain the impact,*	source
95	16	Synopsis	*(at most five in Deep mode, at most three in Fast mode, counting the router call) so that the system runs on free tiers.*	source
99	17	Synopsis	*System Description: the system is modelled as S = {X, Y, C, M, P, F, Sc, Fc}, where C is the set of connectors, M the user's memory set and P the ordered list of language…*	source
101	17	Synopsis	*the conversation (each truncated to 1500 characters) and u the anonymous user identifier.*	source
105	17	Synopsis	□ *Identify data structures, classes, divide and conquer strategies to ex-*	source
110	17	Synopsis	*Functional relations: F = {fplan, fsearch, frank, fsynth, fverify, fpropose, frank : E → fsynth : → and fexecute : (r, approve) →result.*	web + source
119	18	Synopsis	*tence carries a citation to existing evidence, G and the conflicts are reported, every proposal remains pending until the user approves it, and every unavailable source i…*	source

123	18	Synopsis	□ *ACM SIGIR Conference on Research and Development in Information*	source
128	18	Synopsis	□ *IEEE Access (IEEE journal) and the ACL Rolling Review venues*	source
136	19	Synopsis	*A reinforcementlearning reward teaches it when each move pays off.*	source
137	19	Synopsis	*We did not train anything, but we copied the habit of starting from short quotes and capping how many retrieval turns a question gets.*	web + source
141	19	Synopsis	*Cortex keeps the core idea of answering from retrieved text, though our "index" is the user's live Gmail, Slack, GitHub and other accounts, queried at question time.*	web + source
144	19	Synopsis	*Here the model writes a thought, calls a tool, reads the result and repeats.*	source
148	19	Synopsis	*The authors fire a new retrieval after every reasoning sentence.*	web + source
152	19	Synopsis	*A model can answer each part of a question and still get the whole wrong; the paper calls this the compositionality gap and closes some of it by making the follow-up ques…*	source
164	19	Synopsis	*A classifier guesses how hard a question is and picks no retrieval, one step or many steps.*	web + source
165	19	Synopsis	*We hand that choice to the user as the Fast and Deep modes.*	source
171	19	Synopsis	*RARR searches for support after the text is written and edits the parts it cannot back up.*	source
175	20	Synopsis	*Long answers are broken into atomic facts and scored by the share that a knowledge source supports.*	source
176	20	Synopsis	*Our groundedness score is a coarser cousin that works per sentence and only against the evidence that sentence cites.*	source
183	20	Synopsis	*The first teaches a model to call APIs; the second maps the Model Context Protocol ecosystem and its security holes.*	source
188	20	Synopsis	*We planned the work as a waterfall, phase by phase, using the tasks in Section 5.3.1: survey and requirements first, then feasibility, design, coding, testing with evalua…*	source
192	22	Technical Keywords	*Information retrieval and natural language processing: agentic retrievalaugmented generation, multi-source question answering, claim verification (attribution), long-term…*	source
194	22	Technical Keywords	*The keywords follow the 1998 ACM Computing Classification System ( ).*	source
226	23	Technical Keywords	*Additional free keywords: agentic RAG, query decomposition, parallel tool calling, claim verification, groundedness, citation, cross-source conflict detection, explainabl…*	source
231	25	Introduction	*asks a question in plain language, for example "What changed in Orbit this week, and what's at risk?".*	source
236	25	Introduction	□ *The same pipeline feeds a proactive Daily Brief (what changed in the*	source
238	25	Introduction	□ *A fictional sample workspace (a four-person student team building a*	source
243	25	Introduction	*Global Institute, "the average interaction worker spends an estimated 28 percent of the workweek managing e-mail and nearly 20 percent looking for internal information or…*	source
259	26	Introduction	*Requiring a proposal and a separate approval is simple to build and easy to test.*	source
262	26	Introduction	*Generative Agents [13, 14] choose what to remember without asking.*	source
271	26	Introduction	*to-sequence model look things up in an external dense index instead of relying only on what its weights memorised.*	source

277	26	Introduction	*We kept the principle of answering from retrieved text but dropped the single prebuilt index: Cortex searches several live tools through their own APIs and hands the mode…*	source
281	27	Introduction	*what came back, and goes round again; on HotpotQA and FEVER this produced fewer invented facts than reasoning alone.*	source
284	27	Introduction	*[6] noticed that models often get every sub-question right yet still botch the combined question.*	source
288	27	Introduction	*Cortex fits the planning branch of that family: it writes one plan, then runs all of its searches at the same time.*	source
294	27	Introduction	*The reward mixes four things: a correct answer, reading that is grounded, using both search types where each helps, and finishing in few turns.*	source
295	27	Introduction	*On 500 validation questions each from HotpotQA, 2WikiMultiHopQA and MuSiQue they report retrieval recall of 0.90, 0.90 and 0.70, and better answers than every baseline th…*	source
296	27	Introduction	*Looking at the learned behaviour, the authors see the agent search by meaning to land on the right topic, open a paragraph to check details, and then search by keyword fo…*	web + source
297	27	Introduction	*We had no way to train such a policy on hosted free-tier models, so Cortex borrows three of GRASP's lessons as fixed rules instead.*	source
303	27	Introduction	*gle model to fetch passages only when it thinks it needs them and to emit special tokens that rate both the passage and its own answer.*	source
307	28	Introduction	*SearchR1 [22] uses reinforcement learning to teach a model when to issue search queries in the middle of its reasoning.*	source
314	28	Introduction	*RARR [11] takes finished model output, searches for support, and edits away what it cannot support.*	web + source
324	28	Introduction	*In Generative Agents [14], simulated characters log everything they observe and later pull memories back by a weighted score of recency, importance and relevance.*	source
325	28	Introduction	*Cortex's memory service ranks with the same three ingredients (salience for importance, a 30-day half-life for recency, search score for relevance).*	source
329	28	Introduction	*a model teach itself, from its own sampled API calls, which tool to*	source
333	29	Introduction	*[16] map the MCP ecosystem and list the attacks possible at each stage of a server's life, from creation to updates.*	web + source
338	29	Introduction	*inside long prompts and found a U-shaped curve: models do best when the fact is near the start or the end, and worst when it sits in the middle.*	source
340	29	Introduction	*Table 3.1 compares the closest approaches with Cortex on the properties this project targets.*	source
369	31	Problem Definition and scope	*Given a question q from this user, Cortex has a fixed allowance of model calls and tool calls in which to produce an answer.*	web + source
371	31	Problem Definition and scope	*It must also say how much of the answer is supported (c), where the sources disagree and what could not be found (d).*	source
374	31	Problem Definition and scope	*Finally, it should remember only what is likely to matter later, and the user must be able to see and change what it remembers.*	source
379	31	Problem Definition and scope	□ *Input (ask): an HTTP request with a JSON body {}. is a non-empty string of at most 2000 characters; is or (any other value is treated as); is the recent conversation, o…*	source
388	32	Problem Definition and scope	*6 items per call (4 in Fast mode); at most 16 evidence items (8 in Fast mode) reach the model; at most 3 action proposals and 2 memplanner output is validated against a s…*	source

390	32	Problem Definition and scope	*The I/O state behaviour of an action is shown in Figure 6.6.*	source
412	33	Problem Definition and scope	*The Oracle AI Database container could not be used at the time of writing because of a Docker networking issue, so the SQLite fallback is the active backend.*	source
414	33	Problem Definition and scope	*A four-member team within one academic year, alongside regular coursework.*	source
418	33	Problem Definition and scope	*the number of model calls per question, which dominates both latency and free-tier cost, and on other efficiency and quality issues.*	source
421	33	Problem Definition and scope	*The same search repeated within 90 s comes from a separate search cache.*	web + source
422	33	Problem Definition and scope	*Only the synthesis step uses the stronger model tier; the planner and the verifier make do with the faster one.*	source
423	33	Problem Definition and scope	*The lexical verifier is linear in the number of claim and evidence tokens, so we run it first and call a model only for what it cannot settle.*	source
425	33	Problem Definition and scope	*A working web application (Ask with live trace, Brief, Memory, Con-*	source
431	34	Problem Definition and scope	*n (unbounded) Calls grow with the number of steps; hard to keep within free quotas; latency varies widely; can loop.*	source
435	34	Problem Definition and scope	*varies Flexible, but adds a framework dependency and makes the call count less predictable.*	source
451	35	Problem Definition and scope	*A student project team can ask what the guide asked for, which dead-*	source
453	35	Problem Definition and scope	*A developer can check which pull requests wait on them and whether*	web + source
455	35	Problem Definition and scope	*Anyone who starts the day with e-mail can read the Daily Brief instead,*	web + source
457	35	Problem Definition and scope	*A team that needs to audit an assistant can follow each claim back to*	source
458	35	Problem Definition and scope	*its source, and nothing is sent or changed without a person approving it.*	source
472	36	Problem Definition and scope	*RAM 8 GB The Next.js development server, Node.js, the Python service and a browser together need several gigathe optional Oracle Database Free container needs more.*	source
476	36	Problem Definition and scope	*Every model call and live connector call is a remote API request.*	source
481	37	Problem Definition and scope	*Frameworks and libraries: Next.js 16 with React 19 and Tailwind CSS FastAPI, Uvicorn and Pydantic; and (optional); Vitest.*	source
483	37	Problem Definition and scope	*External services (free tiers): Groq, Google Gemini, Cerebras, OpenOracle AI Database (optional).*	source
486	39	Project Plan	*The department's format asks for a waterfall plan, so we followed one.*	source
488	39	Project Plan	*The estimates below use the basic COCOMO model together with the outputs of the laboratory assignments in Annexures B and C.*	web + source
501	39	Project Plan	*With four team members working part-time alongside coursework, this fits the academic year (about ten months including documentation and evaluation).*	source
506	39	Project Plan	*Hardware: the team's laptops (Table 4.2); no dedicated server is*	web + source

524	40	Project Plan	*memory service run as two processes and talk over HTTP/JSON on the same machine.*	source
526	40	Project Plan	*Inside the server, search calls run at the same time as asynchronous promises, while the memory service puts a lock around its SQLite database so that writes do not colli…*	source
527	40	Project Plan	*A few objects exist once per Node.js process (the model router, the action store and the connector registries), and every request shares them.*	source
536	41	Project Plan	*Have top software and customer managers formally committed to support the project?*	source
540	41	Project Plan	*Are end-users enthusiastically committed to the project and the system/product to be built?*	web + source
543	41	Project Plan	*Are requirements fully understood by the software engineering team and its customers?*	source
546	41	Project Plan	*Have customers been involved fully in the definition of requirements?*	source
550	41	Project Plan	*The interface states uncertainty explicitly (struck-through claims, "not found" answers) to keep expectations realistic.*	source
551	41	Project Plan	*agent design, web development and Python are covered, while database operations (Oracle) were new.*	source
554	41	Project Plan	*The functional requirements are stable; external APIs and free-tier limits are not.*	web + source
568	42	Project Plan	*Connector APIs (Composio tool names, MCP servers) change or fail*	source
572	42	Project Plan	*The agent performs an unintended action (e.g. after prompt injection)*	source
587	43	Project Plan	*Medium < 5% Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated*	source
635	46	Project Plan	*Risk Description The Oracle memory backend cannot be brought up.*	source
641	46	Project Plan	*Strategy The memory service offers the same API on SQLite with full-text search and reports the active backend on its health endpoint; Oracle remains optional.*	source
654	47	Project Plan	□ *Task 4: Feasibility spikes: free-tier model failover, Composio OAuth*	web + source
658	47	Project Plan	□ *Task 6: Implementation of the web application: Ask with live trace,*	source
663	47	Project Plan	□ *Task 9: Term-II work: Oracle memory backend, larger evaluation set,*	source
668	48	Project Plan	*The feasibility spikes (T4) ran beside the design so that the riskiest services were tried before coding began.*	source
693	48	Project Plan	*We report to the guide in the project slots of the lab timetable, and progress is noted on the assessment sheet.*	source
698	49	Project Plan	*Composio, MCP, Firecrawl and sample connectors; freetier router, cooldowns and cache*	web + source
701	49	Project Plan	*Ask, Brief, Memory and Connections screens; event stream client; tests and documentation*	source
702	49	Project Plan	*Internal guide Review of design and progress, approval of milestones*	source
705	49	Project Plan	*External guide Industry guidance on behalf of the sponsor, Techtrix Solutions Pvt Ltd*	web + source
713	51	Software requirement specifi	*This chapter pins down what Cortex has to do and under which limits.*	source
719	51	Software requirement specifi	□ *Design the agent pipeline, the connector interface, the model router,*	source

722	51	Software requirement specifi	□ *Write automated tests and an evaluation harness, run them before each*	source
729	51	Software requirement specifi	□ *End user (primary actor): typically a developer, an undergraduate or*	source
730	51	Software requirement specifi	*anyone else whose work is split across Gmail, Google Calendar, Google Drive, Notion, Slack and GitHub.*	source
734	52	Software requirement specifi	□ *Operator (deploys Cortex; may be the same person): places API keys*	source
758	53	Software requirement specifi	*Review the exact preview of a proposed action, optionally edit it, then approve (executes once) or decline*	source
760	53	Software requirement specifi	*Proposal is pending, less than 30 minutes old, same user*	source
774	53	Software requirement specifi	*Declare a server, its search tool and its actions in the configuration file; no code change*	source
808	55	Software requirement specifi	*. . .), source, title, quote (a compact excerpt), and optional link, author, timestamp and sub-question identifier.*	web + source
811	55	Software requirement specifi	*the rewritten question, an optional time window (since, until, label), and up to five sub-questions, each with its sources and a keyword query of at most 80 characters.*	source
816	55	Software requirement specifi	□ *ActionProposal: identifier, connector, action, title, a humanreadable preview, arguments, risk (or), status (, ,,,) and result.*	source
820	55	Software requirement specifi	*"its connection expired; reconnect it") and the impact on the answer.*	web + source
840	56	Software requirement specifi	*Evidence id (PK) source title quote url author timestamp subQuestionId (FK)*	source
864	56	Software requirement specifi	*Figure 6.3 shows Cortex as a single process with its external entities.*	source
868	56	Software requirement specifi	*The response cache and the search cache are omitted for clarity.*	web + source
886	57	Software requirement specifi	□ *Figure 6.5 follows one run from question to stored memory.*	source
888	57	Software requirement specifi	*monds are the four branch points (Fast mode, small talk, no evidence, action requested), and the bars mark where the searches fork and join.*	source
890	57	Software requirement specifi	□ *Interface Requirements: a responsive web interface usable on a lap-*	source
891	57	Software requirement specifi	*top and on a phone; answers streamed as they are written; the reasoning trace available but collapsed by default; a JSON/HTTP and server-sent event interface between brow…*	source
892	57	Software requirement specifi	□ *Performance Requirements: the answer should start streaming within a few seconds, and searches run concurrently.*	source
895	57	Software requirement specifi	*The preliminary evaluation measured a median end-to-end time of 3.7 s and a 90th percentile of 7.1 s in Deep mode.*	source
971	60	Software requirement specifi	*– Availability (related to reliability): when a model provider fails,*	source
972	60	Software requirement specifi	*the router cools it down and moves to the next one.*	source
976	60	Software requirement specifi	*means writing one class that implements the connector interface, or adding one entry to the MCP configuration file.*	web + source

982	60	Software requirement specifi	*– Security: an action can run only through the approval endpoint,*	source
983	60	Software requirement specifi	*only once, and only for the user it was proposed to.*	web + source
992	60	Software requirement specifi	*It is born when the model drafts an action and the store accepts it, and it ends in one of five places: discarded, rejected, expired, executed or failed (double borders).*	source
993	60	Software requirement specifi	*Each arrow is labelled with the event that moves it.*	source
1012	61	Software requirement specifi	*Only free-tier services may be used, which bounds model calls per run*	source
1019	61	Software requirement specifi	*time; a restart discards them, which is safe (nothing executes) but requires asking again.*	web + source
1020	61	Software requirement specifi	*Evidence quotes are compacted to a few hundred characters to keep*	source
1022	61	Software requirement specifi	*Node.js 24 or later and Python 3.11 are required.*	source
1045	62	Software requirement specifi	*Composio SDK / hosted MCP Gmail, Calendar, Drive, Notion, Slack, GitHub tools and OAuth*	source
1048	62	Software requirement specifi	*LangSmith API Optional traces of runs, model calls and searches*	web + source
1053	64	Detailed Design Document usi	*This chapter describes how Cortex is put together: first the architecture, then the data, then the components.*	source
1055	64	Detailed Design Document usi	*The user should see the evidence along with the answer.*	source
1061	64	Detailed Design Document usi	*The browser client (React) renders the Ask, Brief, Memory and Connections screens and consumes the event stream.*	web + source
1063	64	Detailed Design Document usi	*The agent package contains the orchestrator, the prompts, the verifier, the action store, the brief and the tracing wrapper.*	source
1064	64	Detailed Design Document usi	*It depends on three subsystems: the model router package, the connector package and the memory service.*	source
1065	64	Detailed Design Document usi	*A shared package holds the types used by both server and client.*	source
1070	64	Detailed Design Document usi	*The only link between connectors and is that the connectors borrow its cache. stands on its own and is reached over HTTP.*	web + source
1075	64	Detailed Design Document usi	*Deployment to a hosted environment is planned for Term II.*	source
1080	65	Detailed Design Document usi	*Ask + live trace Brief Memory inspector Connections useAgent (SSE client)*	web + source
1118	66	Detailed Design Document usi	*idence without run-scoped identifiers), exchanged between the agent and the connectors.*	source
1122	66	Detailed Design Document usi	*Sections: the synthesis output split at fixed markers into the an-*	source
1145	67	Detailed Design Document usi	*The memory service uses an SQLite database file ( ) with the tables in Table 7.1, and an FTS5 full-text index used for keyword search with BM25 ranking when Oracle is not…*	source
1149	67	Detailed Design Document usi	*The main components are described below; their classes and relationships are shown in Figure 7.2, and the interaction for a consent-gated action in Figure 7.3.*	source

1152	67	Detailed Design Document usi	*chronous generator, enforces the budget, converts every failure into an event, and stamps each step with its duration.*	source
1153	67	Detailed Design Document usi	*Prompts: the planning prompt (returns a JSON plan), the synthe-*	web + source
1156	68	Detailed Design Document usi	*(PK) UUID Owner (workspace-scoped); indexed The remembered text (3–500 characters) priority, preference, fact, person or decision Where it came from (e.g. conversation) W…*	source
1157	68	Detailed Design Document usi	*Conversation; indexed Owner user or assistant Message text (first 4000 characters) ISO timestamp*	source
1160	69	Detailed Design Document usi	*so that both agree on what a claim is), checks citations, overlap and numbers, computes groundedness, and maps conflict lines to evidence.*	source
1172	69	Detailed Design Document usi	*It holds one, one , one and one, and is passed to and, which are plain functions rather than methods.*	source
1174	69	Detailed Design Document usi	*In the connector layer, the registry holds any number of objects, each declaring itss, and five classes implement the interface.*	source
1176	69	Detailed Design Document usi	*Figure 7.3 shows the sequence of messages from a question that asks for an execution happens in a separate request, triggered only by the user.*	source
1226	70	Detailed Design Document usi	*Firecrawl Connector id = web search or scrape a URL; read-only*	source
1252	73	Project Implementation	*Each algorithm was written from the actual code in the repository, so the two can be read side by side.*	web + source
1262	74	Project Implementation	*Validation Zod Schemas for the plan, memory candidates, verification and brief outputs*	source
1279	75	Project Implementation	*emit a failure event (source, reason, impact) for each unavailable*	web + source
1280	75	Project Implementation	*search its working sources again with the first two keywords*	source
1282	76	Project Implementation	*Algorithm 1: Bounded agent run (runAgent), continued: synthesis, verification and proposals*	source
1297	78	Project Implementation	*wait for the provider that recovers soonest if that is within*	source
1300	78	Project Implementation	*invalid key (401/403) →10 min; out of credit (402) → 1 h; model not found (404) →10 min; otherwise*	source
1301	78	Project Implementation	*Streaming uses the same loop, but fails over only before the first token: a stream that breaks after it has started is reported honestly rather than silently restarted on…*	source
1306	79	Project Implementation	*insert m with expiry from its kind (priority 45 d, decision 120 d,*	source
1339	82	Software Testing	*Unit testing (Vitest): pure functions and single classes, such as JSON*	source
1342	82	Software Testing	*the connector registry with a failing connector; and the whole agent run over the sample workspace with a stubbed model, checking the event stream end to end.*	source
1343	82	Software Testing	*System testing and evaluation: the evaluation harness runs 12*	source
1345	82	Software Testing	*Live service checks: a script lists the models of every configured*	source
1347	82	Software Testing	*Interface review: desktop and mobile screenshots of every screen*	source
1354	82	Software Testing	*The evaluation used Deep mode over the fictional sample workspace with free-tier models: Groq as the primary strong model (with as the fast tier) and Gemini 3.5 FlashLite…*	web + source

1358	82	Software Testing	*These results are preliminary: twelve questions over one small fictional workspace are not a benchmark, and repeated runs differ (conflict detection in particular varied …*	source
1360	83	Software Testing	*TL1 Two providers, the first returns HTTP 429; then a second request*	source
1367	83	Software Testing	*TL4 JSON inside a code fence, or embedded in prose*	web + source
1381	83	Software Testing	*TC4 A connector whose search always fails with "fetch failed"*	source
1397	84	Software Testing	*TA4 "Remember that Orbit is my top priority . . . "*	web + source
1408	84	Software Testing	*TA8 Approve by another user; approve by the owner; approve again*	source
1430	85	Software Testing	*In summary, mean groundedness was 0.88 and citation recall 0.92.*	source
1437	87	Results	*The screenshots below show the main screens over the sample workspace.*	source
1438	87	Results	*They were captured for the interface review with the development mock model, which quotes evidence verbatim, so they show the interface and its states rather than the qua…*	web + source
1439	87	Results	*Figure 10.1: Ask screen: a cited answer with a struck-through unsupported claim, a reported conflict between two sources, and the cited sources*	web + source
1447	87	Results	*The lowest values (0.50 and 0.67) came from questions about deadlines ("oa-*	web + source
1448	88	Results	*Figure 10.2: Live trace: step timings, the three-step plan and each tool call with its query, result count and latency*	source
1451	90	Results	*Figure 10.5: Memory inspector (empty state) with the control to teach Cortex a memory of a chosen kind*	web + source
1452	90	Results	*Figure 10.6: Connections: the sample workspace with six sources, and the language model and memory status*	source
1516	92	Results	*The direct question surfaced it, but the broader weekly summary did not, and the behaviour changed from run to run.*	source
1518	92	Results	*Of the two action requests, only the Slack one led to a proposal; the e-mail request produced no Gmail proposal in this run.*	source
1531	94	Deployment and Maintenance	*Optional keys: (own accounts), (web), (tracing), and the Oracle connection settings.*	source
1533	94	Deployment and Maintenance	*Run once to create the Python virtual environment of the memory service.*	source
1535	94	Deployment and Maintenance	*Start the memory service with (port 8000) and, in a second terminal, the web application with ( ).*	source
1542	94	Deployment and Maintenance	*Without any keys, starts a development-only mock model; the application then uses it when is set.*	source
1543	94	Deployment and Maintenance	*Un-installation: stop both processes; delete the repository folder (this removes the dependencies, the virtual environment, the caches in and the memory ledger in); if th…*	web + source
1556	95	Deployment and Maintenance	*Select Trace to see the plan, the searches and their timings.*	web + source
1559	95	Deployment and Maintenance	*the exact content; select Edit to change it, Decline to discard it or Approve to run it once.*	source
1566	95	Deployment and Maintenance	*counts, or connect your own apps with one click (when the server has a Composio key).*	source

1568	95	Deployment and Maintenance	*and what may be missing from the answer, and continues with the others.*	source
1569	95	Deployment and Maintenance	*If every model is busy, wait a minute or switch to Fast mode.*	source
1576	97	Conclusion and future scope	*The answer it streams must cite numbered evidence, and each claim is checked afterwards.*	source
1581	97	Conclusion and future scope	*Over twelve questions the preliminary results back the main design choices.*	web + source
1585	97	Conclusion and future scope	*Conflict detection and action proposals each worked in only one of two cases, and the deadline questions had the lowest groundedness.*	source
1586	97	Conclusion and future scope	*Our base paper, GRASP, shows that both the granularity of evidence and the number of retrieval steps matter.*	source
1590	97	Conclusion and future scope	*by comparing dates and numbers extracted from evidence of different sources about the same entity.*	source
1591	97	Conclusion and future scope	□ *Enlarge the evaluation to the planned 50 questions, repeat runs to*	source
1595	97	Conclusion and future scope	□ *Implement the remaining vision items: time-travel queries over source*	web + source
1597	97	Conclusion and future scope	□ *Expose Cortex itself as an MCP server, so that other assistants can use*	source
1610	108	Annexure Laboratory assignme	*The IDEA matrix (Table B.1) is a grid of verbs, from Increase, Drive, Educate and Accelerate down to Ignore, Decrease, Eliminate and Avoid, that sorts what a product shou…*	source
1611	108	Annexure Laboratory assignme	*The knowledge canvas is a one-page view of the opportunity a product addresses.*	source
1615	108	Annexure Laboratory assignme	*The IDEA matrix applied to Cortex is given in Table B.2, and the knowledge canvas in Table B.3.*	source
1617	108	Annexure Laboratory assignme	*Complete or satisfy ability issues using modern algebra and/or relevant mathematical models.*	source
1627	109	Annexure Laboratory assignme	*Increase Trust in answers: every factual sentence is cited and verified.*	web + source
1636	109	Annexure Laboratory assignme	*Accelerate Repeated questions through response and search caches, and Fast Mode.*	source
1640	110	Annexure Laboratory assignme	*Opportunity Knowledge workers spend a large share of their time looking for information [20]; generative AI targets this work [21].*	source
1642	110	Annexure Laboratory assignme	*Knowledge sources Gmail, Calendar, Drive, Notion, Slack, GitHub, MCP servers, the web; the user's own memories.*	web + source
1643	110	Annexure Laboratory assignme	*Value proposition Cited, verified answers; visible conflicts; explainable memory; actions only with consent; zero cost.*	source
1646	110	Annexure Laboratory assignme	*risk from changing free-tier limits and third-party APIs (Section 5.3 and Table 5.2).*	source
1649	110	Annexure Laboratory assignme	*The problem as implemented is therefore in class P, with model and tool calls treated as bounded oracle calls.*	source
1657	112	Annexure Laboratory assignme	□ *Use of divide and conquer strategies to exploit distributed/parallel/*	source
1658	112	Annexure Laboratory assignme	*concurrent processing of the above to identify object, morphisms, overloading in functions (if any), and functional relations and any other dependencies (as per requireme…*	source

1662	112	Annexure Laboratory assignme	*Morphisms: fsearch maps a (source, query) pair to evidence, fverify maps (answer, evidence) to claims.*	source
1664	112	Annexure Laboratory assignme	*The state behaviour of a proposal is in Figure 6.6.*	source
1698	113	Annexure Laboratory assignme	*mathematical models, GUI, Function testing principles, if any) selection and appropriate use of testing tools, testing of UML diagram's reliability.*	source
1699	113	Annexure Laboratory assignme	*Write also test cases [Black box testing] for each identified functions.*	source
1700	113	Annexure Laboratory assignme	*You can use Mathematica or equivalent open source tool for generating test data.*	source
1703	113	Annexure Laboratory assignme	*Black-box test cases for the identified functions are given in Table C.1; their results are covered by the tests in Chapter 9.*	source
1708	113	Annexure Laboratory assignme	*The references are kept for completeness; the McKinsey reports are also used in the motivation.*	web + source
1712	114	Annexure Laboratory assignme	*Failure with name, reason "could not be reached", impact; run continues*	source
1729	116	Annexure Project Planner	*The project schedule is planned with a Gantt chart (Figure D.1) for the academic year 2026-2027.*	web + source
1730	116	Annexure Project Planner	*Bars show planned durations; the status column shows the state at the time of writing.*	source
1731	116	Annexure Project Planner	*Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr*	source
1753	116	Annexure Project Planner	*T10 Final report and demo Mar 2027 Apr 2027 T9*	source
1758	118	Annexure Reviewers Comments	*(At-least one technical paper must be submitted in Term-I on the project design in the conferences/workshops in IITs, Central Universities or UoP Conferences or equivalen…*	source
1760	118	Annexure Reviewers Comments	*Paper Title: Agentic Retrieval-Augmented Generation over Live Workspace Data: A Survey and the Cortex Assistant*	source
1773	122	Annexure Term-II Project La	*Review of design and necessary corrective actions taking into consideration the feedback report of Term I assessment, and other competitions/conferences participated like…*	web + source
1779	122	Annexure Term-II Project La	*Programming of the project functions, interfaces and GUI (if any) as per 1 st Term term-work submission using corrective actions recommended in Term-I assessment of Term-…*	source
1780	122	Annexure Term-II Project La	*Status: the functions, interfaces and GUI described in this report are implemented; Term-I corrective actions to be applied after the assessment.*	source
1783	122	Annexure Term-II Project La	*Status: Vitest and the evaluation harness are in use (Chapter 9, Figures 10.8 and 10.9); a larger evaluation with repeated runs is planned.*	source

Method, tools and limitations

- **Text extraction:** PyMuPDF 1.28.5 (Python bindings 2.0, Python 3.11.9) from the submitted PDF; hyphenated line breaks joined; running headers, footers and page numbers removed.
- **Matching:** lower-cased word tokens [A-Za-z0-9]+; a word counts as matched if it lies inside any 8-word sequence that also occurs in a source. Shorter shared phrases (technical terms, method names) are not counted, as in most checkers' default "exclude small matches" setting.
- **Sources:** Semantic Scholar Graph API (paper/search/match) for titles and abstracts; arXiv export mirror for full texts; fetched 31 October 2026.
- **Web:** Claude WebSearch tool, one quoted phrase per query (combined OR queries were rejected by the tool).
- **Rendering:** HTML printed to PDF with Google Chrome 156 headless.
- **Scripts and raw data** (extract.py, sentences.py, fetch_sources.py, analyze.py, build_report.py, *_sentences.json, web_log.jsonl, sources.json) are kept with the team for re-running.

Limitations

- No repository check. Turnitin and DrillBit also compare against their stores of student submissions, theses and paywalled journals. This report cannot, so the official index can be higher, mainly from SPPU template wording (lab-assignment and annexure headings) that appears in many earlier BE reports.
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